



Education to Eternity

JAIDEV EDUCATION SOCIETY'S

JD COLLEGE OF ENGINEERING AND MANAGEMENT

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(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATU &amp; RTMNU



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**B. Tech Summer / Winter - 2025**  
**(ESE / Make-up / Supplementary) Examination**

Branch: Computer Science &amp; Engineering

Sem: - V

Subject: PEC-II (React JS)

Subject Code: CS5E002C

Marks: 100

Date of Exam: 11/11/25

Time: 3 Hrs.

**Instructions to the students:**

1. Read the question paper carefully (Branch, Semester, Scheme) before attempting the questions.
2. All the questions carry marks as indicated.
3. Assume suitable data whenever necessary.
4. Illustrate your answer whenever necessary with the help of neat sketches.
5. Use of Programmable Calculator is prohibited.
6. Don't use red pen for writing the answers.

**COs: Students should be able to:**

- CO1: Understand the fundamentals of React and JSX.  
 CO2: Design and develop functional UI components using hooks and props.  
 CO3: Manage application state using use State, use Effect, and Context API.  
 CO4: Integrate APIs and handle routing in React applications.  
 CO5: Build, deploy, and optimize a full-scale single-page React application.

**SECTION A**

1. All Questions are compulsory.
2. Answer in one/two word/sentence.

| Q. No. | Questions                                                                            | Level/CO Mapped | Marks |
|--------|--------------------------------------------------------------------------------------|-----------------|-------|
| Q.1    | Define React components and state their role in UI composition.                      | L1/CO1          | 2     |
| Q.2    | State two advantages of using Virtual DOM over Real DOM.                             | L1/CO1          | 2     |
| Q.3    | Identify the main difference between props and state in React.                       | L2/CO1          | 2     |
| Q.4    | List out four core features offered by React library.                                | L1/CO1          | 2     |
| Q.5    | Give the steps to create a functional component that uses the use State hook.        | L2/CO2          | 2     |
| Q.6    | Classify different types of events supported in React (e.g., keyboard, mouse, form). | L2/CO2          | 2     |
| Q.7    | Restate how the use Effect hook helps simulate component lifecycle behaviour?        | L2/CO2          | 2     |
| Q.8    | Write short notes on the concept of conditional rendering in React.                  | L3/CO3          | 2     |
| Q.9    | Give an example of using Context API for sharing data between multiple components.   | L2/CO3          | 2     |
| Q.10   | Define pagination and state its purpose in data-driven web applications.             | L1/CO4          | 2     |
| Q.11   | State how environment variables help secure API keys in React projects?              | L1/CO5          | 2     |



|      |                                                                                         |        |   |
|------|-----------------------------------------------------------------------------------------|--------|---|
| Q.12 | Enlist any three optimization techniques used to improve React application performance. | L1/CO5 | 2 |
|------|-----------------------------------------------------------------------------------------|--------|---|

## SECTION B

### 1. All Questions are compulsory.

| Q. No. | Questions                                                                           | Level/CO Mapped | Marks |
|--------|-------------------------------------------------------------------------------------|-----------------|-------|
| Q.13   | Distinguish between use State and use Reducer hooks in terms of state management.   | L4/CO2          | 4     |
| Q.14   | Illustrate form validation in React using controlled components.                    | L3/CO2          | 4     |
| Q.15   | Discuss how lifecycle methods in class components are replicated using React hooks. | L2/CO2          | 4     |
| Q.16   | Examine the impact of conditional rendering on user interface performance.          | L4/CO3          | 4     |
| Q.17   | Write short note on route parameters and their usage in React Router v6.            | L3/CO3          | 4     |
| Q.18   | Sketch a component tree diagram showing how props flow through nested components.   | L3/CO3          | 4     |

## SECTION C

### 1. Solve any FOUR Questions.

| Q. No. | Questions                                                                                            | Level/CO Mapped | Marks |
|--------|------------------------------------------------------------------------------------------------------|-----------------|-------|
| Q.19   | Illustrate the working of nested and dynamic routes using React Router v6.                           | L4/CO3          | 7     |
| Q.20   | Explain how custom hooks enhance code reusability and simplify API data fetching?                    | L5/CO4          | 7     |
| Q.21   | Analyse the importance of search and filter features in React data-driven applications.              | L4/CO4          | 7     |
| Q.22   | Examine how Local Storage and Session Storage are used to preserve data state?                       | L4/CO4          | 7     |
| Q.23   | Describe how to organize React components and apply styling using CSS Modules and Styled Components? | L5/CO5          | 7     |
| Q.24   | Discuss how code splitting and lazy loading improve large-scale React project performance?           | L5/CO5          | 7     |

## SECTION D

### 1. Solve any TWO Questions.

| Q. No. | Questions                                                                                                                                     | Level/CO Mapped | Marks |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-------|
| Q.25   | Rewrite about following:<br>a) React Router v6 architecture<br>b) Custom Hook structure<br>c) Nested Routing workflow<br>d) Context API usage | L5/CO3          | 12    |
| Q.26   | Explain the complete process of API integration using Axios /fetch with error and loading handling.                                           | L6/CO4          | 12    |
| Q.27   | Describe the end-to-end workflow of building, optimizing, and deploying a React project on Netlify or Vercel.                                 | L6/CO5          | 12    |